MINOR PROJECT -01

RABBIT PVT. LTD.

MINOR PROJECT ASSESSMENT Network Discovery, Information Gathering, Enumeration and FTP Security Testing Assessment Type Scenario-Based Practical Maximum Marks 50

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Date _29-08-2026_____ Batch / Team __CYBER SECURITY_____ Target Student-selected authorized lab target

SCENARIO

Rabbit Pvt. Ltd. wants a junior security team to perform a short external reconnaissance and enumeration exercise. Students

are not given a target IP in advance. They must first identify a suitable target from Shodan and then perform an authorized,

non-destructive assessment of the selected system. The exercise tests networking knowledge, Nmap usage, service enumeration, FTP investigation, prioritization and security reasoning.

Important: Choose only a target that is clearly intended for security research, a CTF/lab system, your own system, or a host

for which you have explicit permission to test. Do not scan random public organizations or third-party systems.

OBJECTIVE

Starting from an unknown target, demonstrate the complete workflow: identify a suitable authorized target, understand its

network exposure, scan ports, enumerate services, investigate FTP when present, identify a security weakness, and document the reasoning and evidence behind your conclusions.

ASSESSMENT FLOW

Target Selection · Scope · Host Discovery · Port Scanning · Service Detection · Enumeration · FTP Assessment · Security Observation ·

Controlled Validation · Evidence · Report

TASK1 :TARGET IDENTIFICATION:

The target was identified using Shodan as part of the network security assessment. The selected target is a CTF/training system and has been authorized for testing as part of this practical assessment.

1.1 TARGET SELECTION SHEET

Target IP : 40.113.50.46

Hostname : CTF-Win

Organization : Microsoft Corporation

Location : Dublin, Ireland

Target Type : CTF / Training Target

Authorization : Authorized for this practical assessment

Scope : 40.113.50.46 only

1.2 REASON FOR TARGET SELECTION

The target was selected because it is a CTF/training system suitable for practicing network discovery, port scanning, service identification, enumeration, and security assessment techniques. The target provides network-accessible services that can be analyzed as part of the practical exercise.

1.3 AUTHORIZATION

The selected target has been authorized for security testing as part of this practical assessment. The testing will be limited to the specified target and will follow the scope defined for the project.

1.4 SCOPE

In Scope: 40.113.50.46 (CTF-Win)

Out of Scope: All other IP addresses, hosts, domains, and third-party systems.

The assessment will focus on network discovery, TCP port scanning, service/version detection, service enumeration, and FTP assessment if FTP is discovered.

1.5 Shodan Target Discovery Evidence

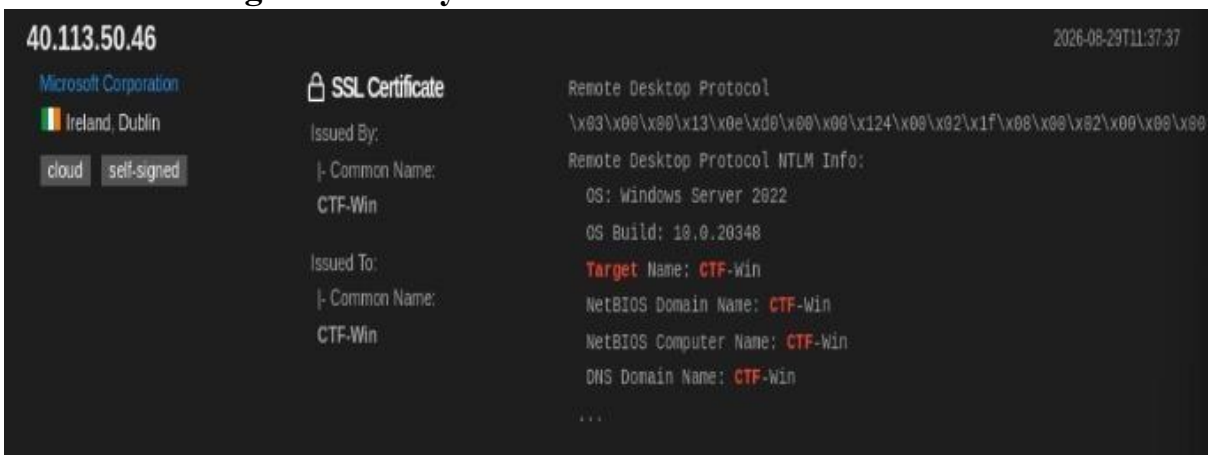


Figure 1: Shodan result showing the selected target 40.113.50.46

TASK 2 : HOST AND PORT DISCOVERY

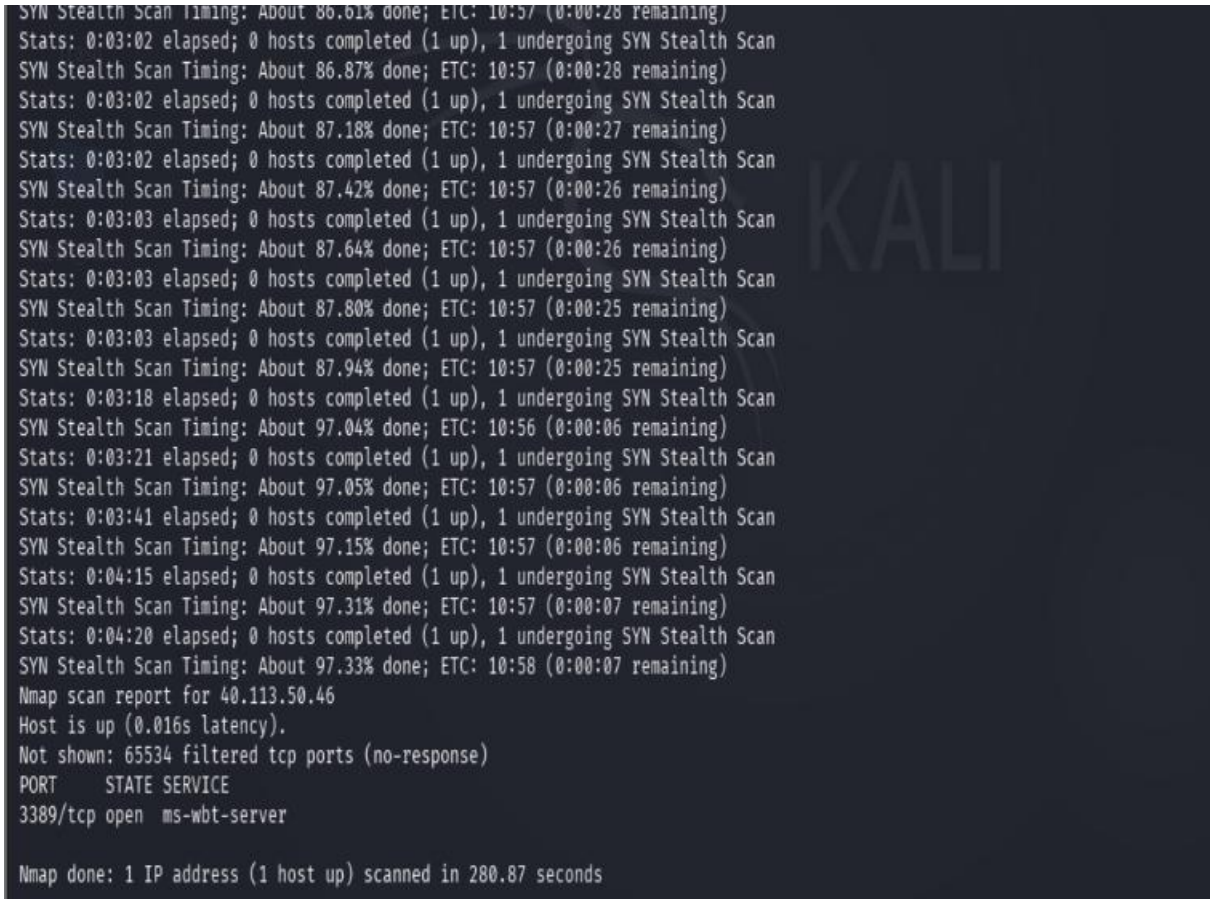
2.1 BASIC NMAP SCAN

A basic Nmap scan was used to quickly identify commonly accessible TCP ports and obtain an initial view of the target's network exposure.



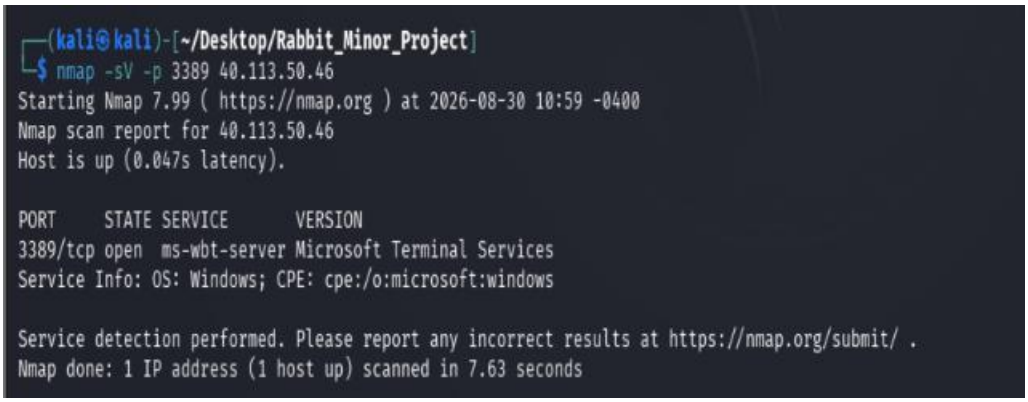
2.2 FULL TCP SCAN

A full TCP scan was performed to examine all TCP ports from 1 to 65535 and ensure that services running on non-standard ports were not missed.



2.3 service/version detection

Service/version detection was selected to determine the software and version associated with the open RDP port**.**



2.4 OS Identification

OS detection was performed to identify the operating-system family and better understand the target environment.



TASK3: BUILD THE ATTACK SURFACE 3.1 service detection

```
┌──(kali㉿kali)-[~/Desktop/Rabbit_Minor_Project]
└─$ nmap -sV -p 3389 40.113.50.46
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-29 11:51 -0400
Nmap scan report for 40.113.50.46
Host is up (0.00073s latency).

PORT     STATE    SERVICE       VERSION
3389/tcp filtered ms-wbt-server

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.23 seconds

┌──(kali㉿kali)-[~/Desktop/Rabbit_Minor_Project]
└─$ nmap -sV -p 3389 40.113.50.46 -oN Nmap/service_version_scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-29 11:52 -0400
Nmap scan report for 40.113.50.46
Host is up (0.043s latency).

PORT     STATE SERVICE       VERSION
3389/tcp open  ms-wbt-server Microsoft Terminal Services
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.20 seconds
```

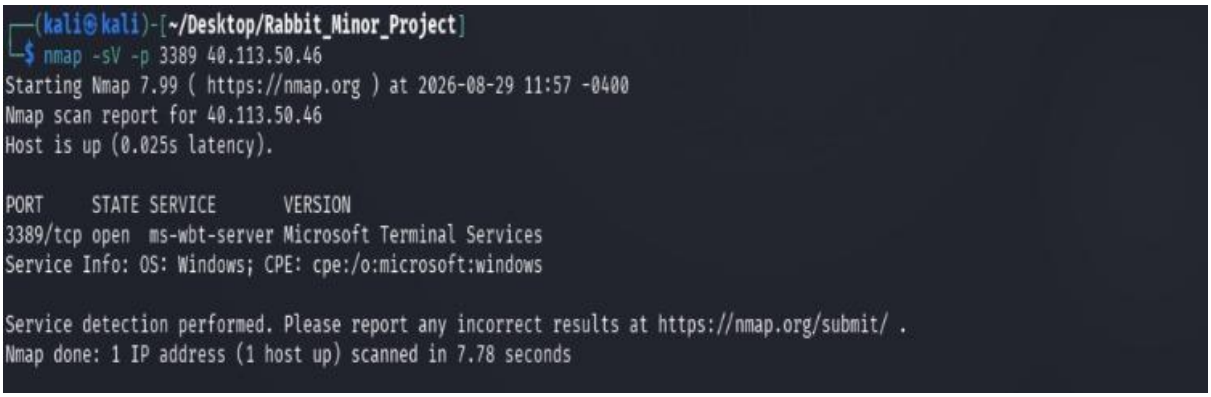
3.2 service enumeration

```
┌──(kali㉿kali)-[~/Desktop/Rabbit_Minor_Project]
└─$ nmap -sV -p 3389 --script rdp-enum-encryption 40.113.50.46
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-29 11:54 -0400
Nmap scan report for 40.113.50.46
Host is up (0.042s latency).

PORT     STATE SERVICE       VERSION
3389/tcp open  ms-wbt-server Microsoft Terminal Services
| rdp-enum-encryption:
|   Security layer
|     CredSSP (NLA): SUCCESS
|     CredSSP with Early User Auth: SUCCESS
|_    RDSTLS: SUCCESS
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.85 seconds
```

3.3 unknown service investigation



3.4 strategic question

If I find five different services but have limited time, I would first look at the service that could have the biggest security impact. I would start with SMB on port 445 because it is used for file and resource sharing and, if it is not properly secured, it could expose important information or allow unauthorized access. After that, I would check the other services based on how exposed they are and how serious their potential security risks are.